

# Robot / rover

DO

OUTPUT

MOVEMENT

## What does it do?

A two-wheel rover uses a separate geared motor for each wheel. A caster or smooth skid supports the other end.

Drive both wheels forwards to move; change their speeds or directions to turn. A motor driver handles their power.

The Robot class joins the two motors into one object. This example uses a breadboard-mounted L293D chip and a separate motor battery pack.

## Uses

### Robot vacuum cleaner



**DETECT**  
Sense a  
nearby  
obstacle

**DECIDE**  
Is the path  
clear?

**DO**  
Drive on or  
turn away

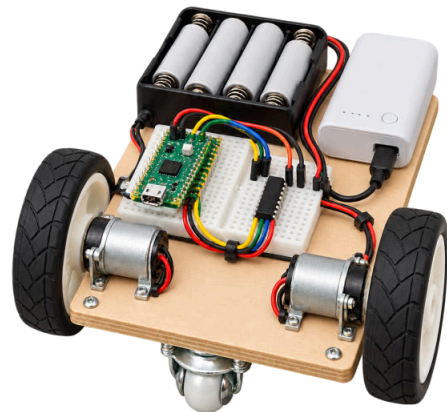
### Sock delivery turtle



**DETECT**  
Read delivery  
button

**DECIDE**  
Is a delivery  
requested?

**DO**  
Drive the sock  
parcel



#### Drive wheels

One  
geared  
motor per  
side

#### Motor driver

L293D, a  
16-pin  
breadboard  
chip

#### Caster or skid

Supports  
the  
chassis

#### Power

Separate  
motor  
and USB  
supplies

# Code

## ● Raspberry Pi

```
from gpiozero import Motor, Robot
from time import sleep

robot = Robot(left=Motor(13, 19),
              right=Motor(17, 27))
try:
    robot.forward()
    sleep(1)
finally:
    robot.stop()
```

## ● Raspberry Pi Pico

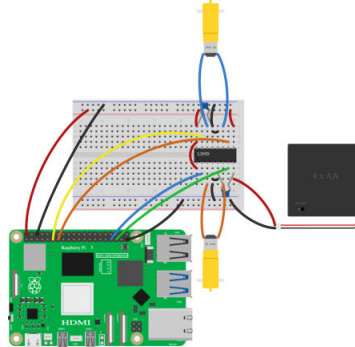
```
from picozero import Robot

robot = Robot(left=(14, 15),
              right=(22, 21))
robot.forward(t=1, wait=True)
```

Test with wheels lifted. Reverse a motor by swapping its leads with power off. Try left(), right(), backward() and stop().

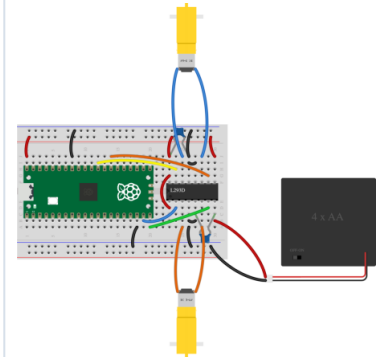
# Wiring

## RASPBERRY PI 5



**GPIO13/19** → L293D pins 2/7;  
**GPIO17/27** → 15/10. **5 V** → 1, 9, 16. GND  
→ 4, 5, 12, 13.

## PICO



**GP14/15** → pins 2/7; **GP22/21** → 15/10.  
**VBUS** → 1, 9, 16. **4×AA alkaline (6 V)** →  
8/GND; GND shared. Two 100 nF  
capacitors. 6 V motors: stall current **≤400**  
**mA each.**